

Forest: A data analysis library for raw high-throughput digital phenotyping data

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Software

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Summary

Forest is a data analysis library designed for raw, high-throughput digital phenotyping data. It is intended to integrate directly with its companion software—the Beiwe platform—which enables large-scale data collection from both Android and iOS smartphones. Beiwe supports digital phenotyping by gathering both active data (such as surveys and audio samples) and passive data (such as accelerometer data) from participants' smartphones. Most smartphone applications depend on software development kits (SDKs) that generate behavioral summary measures using closed, proprietary algorithms that lack external validation. As a result, these applications often fall short of scientific standards for reproducibility and may force researchers to adjust their scientific questions based on the constraints of available data summaries. By contrast, Beiwe collects raw sensor and phone usage data, with collection parameters that can be customized to match the needs of any scientific study.

The analysis of such raw data however presents significant challenges, requiring dedicated statistical methods and specialized software. This is the need that Forest addresses. While the quantitative methods necessary for analyzing raw digital phenotyping data have been developed and published elsewhere, Forest provides robust, accessible software implementations of these methods in Python. Each method within Forest is named after a tree—for example, Jasmine implements sparse online Gaussian process imputation for missing GPS trajectories. We expect Forest to continue growing as additional methods become available.

Statement of need

Background. The phenotype of an organism comprises a collection of traits, such as enzyme activity, hormone levels, and behavior. Increasingly, researchers advocate for a more substantial role for large-scale phenotyping as a route to advances in the biomedical sciences ([Bilder et al., 2009](#); [Delude, 2015](#); [Houle et al., 2010](#); [Robinson, 2012](#)). Of the many types of phenotypes, social, behavioral, and cognitive phenotypes are particularly challenging to study due to their temporal and contextual dependencies. Traditionally, various surveys and assessments are used to ascertain these phenotypes, but these are cross-sectional, subjective, and often burdensome. The ubiquity and capabilities of smartphones—when coupled with appropriate data analytic

techniques—can help overcome these limitations. We coined the term digital phenotyping to refer to the “moment-by-moment quantification of the individual-level human phenotype in situ using data from personal digital devices, in particular smartphones” (Onnela & Rauch, 2016; Torous et al., 2016). Onnela Lab began developing Beiwe, an open source high-throughput smartphone-based digital phenotyping platform that collects active and passive data, with support from an NIH Director’s New Innovator Award, in 2013. The platform consists of iOS and Android front-end applications and an AWS-based back-end infrastructure with a web-based study management portal for data processing and secure storage. This approach not only allows for more objective measurement of known phenotypes but can also give rise to entirely new phenotypes.

State of the field. Social and behavioral phenotypes have traditionally been ascertained using either participant-administered or investigator-administered surveys and assessments in research settings and patient-reported or clinician-reported equivalents in clinical settings. For example, the Amyotrophic Lateral Sclerosis Functional Rating Scale - Revised (ALSFRS-R) includes 12 items (questions), each scored on a 0 (no function) to 4 (full function) scale, and has been used both for diagnosing patients and for tracking disease progression (Mora, 2017). In observational studies and clinical trials, it may be administered every six weeks with smaller within-subject standard deviation when administered by the participant themselves rather than when administered by a clinician (Berry et al., 2019). In order to eliminate recall bias, some of the items in ALSFRS-R can potentially be measured objectively in real-world settings. For example, two items in the ALSFRS-R relate to physical activity: walking (Item 8) and climbing stairs (Item 9). Both of these can be estimated using smartphone accelerometer and gyroscope data (Strackiewicz et al., 2021).

An important development in this field has been the introduction of software development kits (SDKs) for smartphones, such as Apple’s ResearchKit and Google’s ResearchStack, which have facilitated the creation of software for these devices. However, the use of prepackaged software restricts the types of data that can be collected, thereby limiting the research questions that can be investigated and the data analyses that can be performed (Onnela, 2021). For example, Apple’s ResearchKit does not support background sensor data collection (ResearchKit, 2018); Apple’s HealthKit does support background data collection for selected sensors only (Apple Developer, 2020b); and the Core Motion framework allows the collection of raw accelerometer data in the background, but only for up to 12 hours at a time (Apple Developer, 2020a). The algorithms underlying HealthKit metrics, such as step count (Apple Developer, 2020c), are proprietary and subject to change without notice. The use of closed algorithms, which may be updated at any time, makes it hard or impossible to compare data collected at different times or across different SDKs.

Forest appears to be the only open-source library for the analysis of raw, high-throughput smartphone data. Most investigators currently rely on data from commercially maintained SDKs as described above. The small number of investigators who use raw data often apply general-purpose statistical methods not designed for this type of data, or ad hoc methods whose statistical properties are not well understood. Having statisticians, data scientists, and machine learning experts develop appropriate methods and implement them in accessible software like Forest is highly beneficial. This is expected to improve both the quality of statistical analyses and the scientific evidence generated.

State of the Field

Forest operates within a growing ecosystem of tools for mobile health and digital phenotyping. Existing approaches broadly fall into two categories: (1) smartphone software development kits (SDKs) such as Apple’s ResearchKit and Google’s ResearchStack, and (2) general-purpose data science tools (e.g., Python and R libraries for time series and signal processing). SDK-based approaches facilitate data collection but typically provide precomputed behavioral summary

92 measures derived from proprietary algorithms that are not externally validated and may change
93 over time. This limits reproducibility and constrains downstream scientific analysis.

94 In contrast, general-purpose tools offer flexibility but lack domain-specific methods for handling
95 high-frequency, irregular, and missing smartphone sensor data. As a result, researchers often
96 implement ad hoc pipelines, leading to inconsistent methodologies and unclear statistical
97 properties. To our knowledge, there are few open-source libraries that provide standardized,
98 statistically grounded methods specifically for raw digital phenotyping data collected from
99 smartphones.

100 Forest addresses this gap by providing an open-source, domain-specific analysis library designed
101 explicitly for raw digital phenotyping data. Rather than duplicating existing SDK functionality,
102 Forest complements data collection platforms like Beiwe by focusing on statistically principled
103 transformation of raw data into interpretable features. The decision to develop a new library
104 reflects the need for tightly integrated methods that account for the unique structure, scale,
105 and missingness patterns of smartphone-derived behavioral data.

106 Software Design

107 Forest is designed as a modular analysis library organized around independent processing units
108 referred to as “trees,” each implementing a specific methodological pipeline (e.g., mobility
109 inference, accelerometry processing, communication analysis). This architecture reflects a
110 trade-off between extensibility and cohesion: individual trees can be developed and validated
111 independently while adhering to shared data structures and conventions provided by the Poplar
112 utility layer.

113 A key design decision is the separation between data collection (handled by the Beiwe platform)
114 and data analysis (handled by Forest). This improves reproducibility by ensuring that raw
115 data remain unchanged and that all derived features are generated through transparent,
116 version-controlled methods. Forest operates directly on raw data files rather than requiring
117 intermediate proprietary formats, enabling full auditability of the analysis pipeline.

118 The library is implemented in Python to leverage its scientific computing ecosystem and
119 accessibility to researchers. Design choices also prioritize scalability and robustness to missing
120 data, which is pervasive in smartphone sensing. Methods such as Jasmine explicitly model
121 missingness using probabilistic approaches rather than relying on simple interpolation.

122 Research Impact Statement

123 Forest has been developed in conjunction with the Beiwe platform, which has been used
124 in observational studies and clinical research settings across multiple domains, including
125 psychiatry, neurology, and oncology. The analytical methods implemented in Forest—such as
126 mobility trajectory inference from GPS data and accelerometer-based estimation of physical
127 activity—are grounded in peer-reviewed statistical research and have been applied in published
128 studies examining behavioral and clinical outcomes.

129 By providing standardized, open-source implementations of these methods, Forest enables
130 reproducible analysis of high-throughput digital phenotyping data, addressing a key limitation
131 of existing approaches that rely on proprietary or ad hoc analytical pipelines. The library
132 supports large-scale datasets and facilitates the generation of consistent summary metrics
133 across studies, improving comparability and enabling downstream statistical analysis.

134 Forest also serves as a foundation for ongoing methodological development. Its modular design
135 allows new analytical methods to be incorporated as additional trees, supporting the continued
136 evolution of digital phenotyping research. In addition, the inclusion of synthetic data generation

137 tools enables testing, benchmarking, and validation of analytical pipelines, further supporting
138 reproducible and transparent research practices.

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144 AI Usage Disclosure

145 Generative AI tools were used in a very limited capacity to assist with minor editing and
146 refinement of the manuscript. The authors wrote the paper in full, and all content, including any
147 AI-assisted edits, was carefully reviewed and verified for accuracy, consistency, and alignment
148 with the underlying methods and software implementation.

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